Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an On-device AI GPS Tracker for Neutralizing Jamming and Spoofing Attacks in North American Logistics**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		***Mechanical・Materials***	
	Medium Category	Small Category	Medium Category	Small Category
	Distribution / Logistics / Marketing Services	***Intelligent Enterprise Logistics Support technology***	***Automobile / Rail Vehicles***	***Safety Improvement Technologies***
Concept of Proposed Technology	**Core Concept** The objective of this project is to fundamentally resolve cargo theft and security breaches caused by the technical vulnerabilities of existing GPS trackers in North America. By developing an On-device AI GPS Tracker where the AI independently identifies threats, we aim to ensure real-time security visibility and prevent economic losses even under sophisticated jamming or spoofing attacks. **Key Functions and Potential Applications** ● Integration of Server-Independent On-device AI: This project implements an AI engine that detects threats directly within the device. By embedding Purdue's Direct-AI, it analyzes the			

	"Signal DNA" to identify attacks with a 70%+ Recall rate without server dependency. ● Real-time Standalone Defense Architecture: Even during total communication blackouts, the internal AI performs Real-time threat judgment. Upon detection, the device triggers local emergency protocols and data logging, maintaining security visibility regardless of network status. ● Paradigm Shift to Active On-site Defense: This technology transitions logistics security from passive post-incident tracking to active real-time defense, providing the highest level of security reliability and significantly reducing asset loss in the North American market.
Rationale for Joint R&D Project	**Critical Technical Barriers** Legacy GPS tracker has limitations of Mid-range and lower-end GPS tracking with a 70% market penetration rate. ● Security Blackouts during Communication Failures: Most trackers currently deployed rely on passive transmission to external servers. If criminals use jamming to block communication, the device immediately loses its connection, leading to a total Security Blackout where tracking becomes impossible. ● Vulnerability to Spoofing: Legacy devices lack the computational power to verify the physical integrity of GSP signals. Consequently, they are defenseless against spoofing attacks that manipulate routes, often reporting false locations while the actual cargo is being stolen. ● Reactive Post-Incident Handling: Due to these vulnerabilities, logistics firms can only respond after an incident has occurred, leading to an annual loss exposure of $35 billion. **Why Global Collaboration?** This project aims to become a game changer in the North American logistics market through the strategic combination of Korean SME, which possesses development capabilities, and Purdue University, a global leader in GNSS security technology. ● Lead Organization's Proven Commercialization Capabilities: The Lead Organization holds core technologies for specialized communication terminals and has an extensive track record in supplying devices to major telecommunications carriers. Our expertise in the entire lifecycle from design to global commercialization ensures that the final product meets the rigorous operational demands of the North American environment. ● Acquisition of High-Quality Data for AI Modeling: The performance of On-device AI (70%+ Recall) relies heavily on high-fidelity training data reflecting actual attack scenarios. Purdue University provides access to its extensive datasets of North American jamming and spoofing attacks, collected over years via PURT/PUC facilities. Access to this real-world, high-quality data which is unattainable through domestic research is a critical catalyst for achieving global standard precision in our AI models. ● Purdue's Direct-AI Expertise and Track Records: Purdue holds proprietary records in Edge Inference Optimization and adaptive anti-jamming algorithms proven through U.S. DoD projects. Integrating their advanced intelligence and data with our manufacturing excellence is the key to market success.

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	● Achieving Success through Exclusive Infrastructure (PURT/PUC): Utilizing Purdue's exclusive facilities (PURT* and PUC**) allows us to objectively certify a 70%+ Recall rate during communication blackouts, providing the technical credibility essential to secure contracts with major U.S. trucking fleets. * PURT (Purdue University UAS Research and Test Facility) is the largest indoor motion-capture facility in the world dedicated to Unmanned Aerial Systems (UAS) and autonomous vehicle research. ** PUC (Purdue Urban Canyon) is a unique outdoor laboratory designed to replicate the "Urban Canyon" effect found in major North American cities (e.g., New York, Chicago, or Los Angeles)
Key Words	1. Jamming 2. Spoofing 3. On device 4. Direct-AI 5. GPS Tracker

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

The objective is to secure core technologies for an On-device AI GPS Tracker that independently performs asset protection during jamming or spoofing-induced blackouts.

Detailed Objectives

- Development of On-device AI Security Engine: Optimizing lightweight AI algorithms to achieve a 70%+ Recall rate against jamming and spoofing without server dependency.
- Standalone Real-time Response Architecture: Establishing a real-time security framework that triggers local emergency protocols and data logging upon threat detection.
- Global Reliability Validation: Verifying operational stability through North American infrastructure (PURT/PUC) and 10,000km of field operational tests.

R&D Target Specifications

Key Performance Indicator (KPI)	Target Value	Competitive Advantage	Verification Method
Detection Recall	≥ 70%	Improved from <10%(legacy device) for attacks	Validation at PURT/PUC
Detection Latency	Real-time	Shift from reactive (post-incident) to proactive	On-device processing time
Operational Availability	Standalone Mode	Overcomes blackouts during jamming	Communication cut-off simulation

R&D Plan

Through organic collaboration with Purdue University, this project follows a 3-step roadmap: "Data Acquisition → AI Implementation → Field Validation."

This ensures the development of a practical security solution optimized for the North American logistics environment.

- Step 1: Acquisition of N. American Attack Datasets & HW Design (Year 1)
  - Data Acquisition: Constructing AI training datasets by acquiring real-world jamming and spoofing signal data from Purdue's PURT/PUC facilities. This high-fidelity data, unattainable through domestic research alone, serves as the foundation for the AI model.
  - HW Design: Designing a high-reliability MCU-based hardware platform and developing prototypes to support on-device inference.
- Step 2: On-device AI Implementation & Performance Optimization (Year 2)
  - AI Implementation: Optimizing Direct-AI algorithms for the terminal environment (Lightweight Inference) using the acquired datasets. Developing a standalone security engine that functions without server connectivity.
  - Verification: Utilizing Purdue's PURT/PUC to verify the 70%+ Recall rate during

	blackouts and iteratively refining the model to maintain FAR ≤ 1.0%. ● Step 3: N. American Field Operational Test (FOT) & Launch (Year 3) ■ Field Testing and Validation: Conducting 10,000km of real-world road tests across North American logistics routes to confirm long-term operational stability and security efficacy.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Purdue University: Core AI Technology & Scientific Validation ■ High-Fidelity Dataset & AI Modeling : Providing proprietary North American attack datasets and refining the AI algorithms to ensure a Recall rate of 70% or higher. ■ Infrastructure-based Performance Verification : Conducting rigorous security tests using PURT and PUC facilities to provide objective, third-party validation of the tracker's efficacy. ■ Technical Consulting & Standardization : Offering guidance on global GNSS security trends and providing the technical authority required for contracts with major U.S. logistics fleets.
Expected R&R of Korean SME Partner	● Korean SME: Hardware Design, Validation Infrastructure & Commercialization ■ On-device AI Hardware Design : Designing and manufacturing a high-reliability hardware platform capable of on-device AI inference, meeting the rigorous standards of the North American logistics environment. ■ Establishment of Validation Infrastructure : · Control Server: Developing a specialized monitoring and control server to track real-time attack scenarios and device response statuses during testing. · Data Logger: Implementing a high-performance data logging system to preserve AI inference logs and raw GPS data during communication blackouts for forensic analysis. ■ System Integration & Commercialization: Integrating the optimized AI engine into the firmware, acquiring global certifications (e.g., FCC), and finalizing the mass-production design for market entry.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Economic: Preventing massive asset losses and providing a scalable, low-cost security solution for global fleets. ● Technical: Setting a new global standard for active on-device defense, securing a dominant position in the North American logistics security market.

6. References